Zero-Coupon Treasury PCA and Volatility Analysis

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Introduction

This report analyzes the dynamics of U.S. Treasury zero-coupon rates, principal-component bond volatility, and S&P volatility. The analysis focuses on annual data over the available sample period, primarily 1962–2025.

Problem 1: AR(1) Model for PC1 Bond Volatility

Method

Let V_t denote the annual volatility of the first principal component obtained from the log zero-coupon Treasury rates. The logarithm of volatility is modeled using an AR(1) process:

$$\ln V_t = \alpha + \beta \ln V_{t-1} + Z_t.$$

The model was estimated using annual PC1 bond volatility for 1962–2025.

Results

The estimated model is

$$\boxed{\ln V_t = -0.4810 + 0.8550 \ln V_{t-1} + Z_t}$$

An important detail is that I used the "arima () function" and R's `arima()` function reports the intercept as the unconditional process mean rather than directly reporting α in the equation above. The reported process mean was -3.3168. Therefore,

$$\alpha = -3.3168(1 - 0.8550) \approx -0.4810.$$

Residual Diagnostics

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.01056	Reject normality
Jarque-Bera	0.002474	Reject normality

Diagnostic	p-value	Decision at 5%
Ljung-Box Z_t, lag 5	0.02538	Reject no autocorrelation
Ljung-Box Z_t, lag 10	0.09579	Fail to reject
Ljung-Box $\|Z_t\|$, lag 5	0.4226	Fail to reject
Ljung-Box $\|Z_t\|$, lag 10	0.6600	Fail to reject

The ACF of Z_t does not display large individual autocorrelation spikes, while the ACF of $|Z_t|$ also does so. However, the formal Ljung-Box test detects joint residual autocorrelation through lag 5.

The QQ plot shows departures from the normal reference line. This is supported by both formal normality tests.

Conclusion

The residual diagnostics do not support the assumption that Z_t is IID Gaussian. Both the Shapiro-Wilk and Jarque-Bera tests reject normality. The Ljung-Box test also detects residual autocorrelation through lag 5, although this evidence is not significant through lag 10. No significant dependence is detected in the absolute residuals. The plots thou look like white noise.

Problem 2: VAR(1) Model for Bond and S&P Volatility

Method

The next model examines bond and S&P volatility jointly.

Let

$$\ln V_{B,t}$$

denote the logarithm of PC1 bond volatility and

$$\ln V_{S,t}$$

denote the logarithm of S&P volatility.

A VAR(1) model was estimated:

$$\ln V_{B,t} = a_1 + b_{11} \ln V_{B,t-1} + b_{12} \ln V_{S,t-1} + Z_{1t},$$

and

$$\ln V_{S,t} = a_2 + b_{21} \ln V_{B,t-1} + b_{22} \ln V_{S,t-1} + Z_{2t}.$$

Results

The estimated bond-volatility equation is

$$\boxed{\ln V_{B,t} = -0.67543 + 0.83283 \ln V_{B,t-1} + 0.06731 \ln V_{S,t-1} + Z_{1t}}$$

The estimated S&P-volatility equation is

$$\boxed{\ln V_{S,t} = 1.59852 + 0.06986 \ln V_{B,t-1} + 0.36654 \ln V_{S,t-1} + Z_{2t}}$$

Diagnostics for Bond Innovations Z_{1t}

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.01165	Reject normality
Jarque-Bera	0.002303	Reject normality
Ljung-Box Z_1, lag 5	0.03437	Reject no autocorrelation
Ljung-Box Z_1, lag 10	0.09572	Fail to reject
Ljung-Box $\|Z_1\|$, lag 5	0.4751	Fail to reject
Ljung-Box $\|Z_1\|$, lag 10	0.6671	Fail to reject

The bond-volatility innovations therefore continue to show evidence against IID Gaussian behavior.

Diagnostics for S&P Innovations Z_{2t}

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.08329	Fail to reject normality
Jarque-Bera	0.2032	Fail to reject normality
Ljung-Box Z_2, lag 5	0.7871	Fail to reject
Ljung-Box Z_2, lag 10	0.8667	Fail to reject
Ljung-Box $\|Z_2\|$, lag 5	0.9181	Fail to reject
Ljung-Box $\|Z_2\|$, lag 10	0.9455	Fail to reject

The diagnostics for the S&P innovations are consistent with IID Gaussian behavior.

Conclusion

The two innovation series behave differently. The bond innovation Z_{1t} is not consistent with IID Gaussian behavior.

In contrast, normality is not rejected for the S&P innovation Z_{2t}, and there is no significant autocorrelation in either Z_{2t} or $|Z_{2t}|$. The diagnostics for Z_{2t} is consistent with IID Gaussian behavior.

Problem 3: Annual Return on a 7-Year Zero-Coupon Treasury Bond

Method

For continuously compounded zero-coupon rates, the price of an n-year zero-coupon bond is

$$P_n(t) = e^{-nR_n(t)},$$

where $R_n(t)$ is the continuously compounded n-year zero-coupon rate.

A 7-year zero-coupon bond purchased at the end of year t becomes a 6-year zero-coupon bond at the end of year $t+1$.

Therefore, its one-year continuously compounded return is

$$r_{t+1} = -\ln P_7(t) + \ln P_6(t+1).$$

Substituting the zero-coupon pricing formula, we get

$$\boxed{r_{t+1} = 7R_7(t) - 6R_6(t+1)}$$

The last available Treasury observation in each calendar year was used. Since the rates in the source data are reported in percentage points, I divided them by 100 before calculating returns.

Example Results

The first several annual log returns are:

Year	Log Return
1962	0.056351
1963	0.017093
1964	0.037941
1965	0.001551
1966	0.050432
1967	-0.012049

Here, the 1962 return represents the return from purchasing a 7-year zero-coupon Treasury bond at the end of 1961 and valuing it as a 6-year bond at the end of 1962.

Missing-Data Issue

A significant data issue occurs in 1992. Both the year-end 6-year and 7-year zero-coupon rates are missing.

Consequently,

$$\boxed{r_{1992} = NA, \qquad r_{1993} = NA.}$$

I deliberately retaiend the missing returns because removing 1992 before constructing returns would incorrectly connect 1991 directly to 1993 and treat a two-year interval as a one-year holding period.

Conclusion

The missing 1992 Treasury observations necessarily result in missing returns for both 1992 and 1993.

Problem 4: AR(1) Model for the 7-Year Zero-Coupon Rate

Method

Let R_t denote the 7-year zero-coupon Treasury rate at the end of year t.

The model is

$$R_t = a + bR_{t-1} + Z_t.$$

Because the 1992 observation is missing, simply deleting the missing observation before creating the lag would incorrectly use the 1991 rate as the lag for 1993.

Instead, the current and lagged observations were constructed while preserving the calendar-year structure. Any observation for which either R_t or the true R_{t-1} was unavailable was excluded from estimation.

Results

The estimated model is

$$\boxed{R_t = 0.004528 + 0.919397R_{t-1} + Z_t}$$

The model has an R^2 of approximately 0.847.

Residual Diagnostics

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.7708	Fail to reject normality
Jarque-Bera	0.6804	Fail to reject normality
Ljung-Box Z_t, lag 5	0.8980	Fail to reject
Ljung-Box Z_t, lag 10	0.9012	Fail to reject
Ljung-Box $\|Z_t\|$, lag 5	0.07294	Fail to reject
Ljung-Box $\|Z_t\|$, lag 10	0.3606	Fail to reject

Conclusion

The QQ plot is reasonably close to the normal reference line, and both formal normality tests fail to reject normality.

The residuals are consistent with normality, and there is no statistically significant autocorrelation in either the residuals or their absolute values at lags 5 or 10. Therefore, the residual diagnostics are consistent with the assumption that the innovations are IID Gaussian.

Problem 5: 7-Year Rate Model with PC1 Bond Volatility

Method

The 7-year Treasury rate was next modeled as a function of its previous value and the PC1 bond-volatility measure:

$$R_t = a + bR_{t-1} + cV_t + V_tW_t.$$

Here, V_t is the annual PC1 bond volatility used in Problem 1.

The error term is scaled by V_t. Dividing the entire model by V_t gives

$$\frac{R_t}{V_t} = a\frac{1}{V_t} + b\frac{R_{t-1}}{V_t} + c + W_t.$$

Missing-Data and Lag-Construction Challenge

An important implementation issue occurred when the Treasury-rate data were merged with the PC1-volatility data.

I did same thing as problem 4

Results

The transformed regression produced

$$a = 0.009511, \qquad b = 0.896185, \qquad c = -0.067769.$$

Therefore, the fitted model in its original form is

$$\boxed{R_t = 0.009511 + 0.896185R_{t-1} - 0.067769V_t + V_tW_t}$$

Residual Diagnostics

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.1495	Fail to reject normality
Jarque-Bera	0.9604	Fail to reject normality
Ljung-Box W_t, lag 5	0.8501	Fail to reject
Ljung-Box W_t, lag 10	0.5443	Fail to reject
Ljung-Box $\lvert W_t\rvert$, lag 5	1.137×10^{-11}	Reject
Ljung-Box $\lvert W_t\rvert$, lag 10	1.876×10^{-14}	Reject

The QQ plot shows some departures from the normal reference line, particularly in the tails. However, neither formal normality test rejects normality at the 5% significance level.

There is also no statistically significant autocorrelation in W_t itself.

Conclusion

The innovations W_t are consistent with normality, and there is no significant direct autocorrelation in W_t. However, there is very strong autocorrelation in $|W_t|$.

Therefore,

$$\boxed{W_t \text{ is not consistent with an IID Gaussian process.}}$$

Problem 6: Log 7-Year Rate Model with PC1 Bond Volatility

Method

Problem 6 repeats Problem 5 using the natural log of the 7-year zero-coupon rate instead of the level of the rate.

The model becomes

$$\ln R_t = a + b \ln R_{t-1} + cV_t + V_t W_t.$$

Dividing by V_t gives

$$\frac{\ln R_t}{V_t} = a\frac{1}{V_t} + b\frac{\ln R_{t-1}}{V_t} + c + W_t.$$

The same calendar-year lag construction used in Problem 5 was retained..

Results

The transformed regression gives

$$a = -0.27206, \qquad b = 0.88190, \qquad c = -1.45612.$$

Therefore, the estimated original model is

$$\boxed{\ln R_t = -0.27206 + 0.88190 \ln R_{t-1} - 1.45612V_t + V_t W_t}$$

Residual Diagnostics

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.5650	Fail to reject normality
Jarque-Bera	0.5009	Fail to reject normality
Ljung-Box W_t, lag 5	0.5535	Fail to reject
Ljung-Box W_t, lag 10	0.5874	Fail to reject
Ljung-Box $\|W_t\|$, lag 5	0.2762	Fail to reject
Ljung-Box $\|W_t\|$, lag 10	0.5333	Fail to reject

The QQ plot is reasonably consistent with a normal distribution. Both formal normality tests fail to reject normality.

The ACF of W_t is consistent with white noise, and the Ljung-Box tests find no significant autocorrelation through either lag 5 or lag 10.

So unlike Problem 5, the absolute innovations also show no statistically significant serial dependence.

Conclusion

The diagnostics are consistent with the assumption that

$$\boxed{W_t \text{ is IID Gaussian.}}$$

WE can conclude that the log transformation of the 7-year rate therefore substantially improves the innovation diagnostics.

After applying yeo Johnson transform to 1 and 2

Because the original residual diagnostics for Problems 1 and 2 did not fully support the IID Gaussian assumption, a Yeo-Johnson transformation was applied to check whether the residual behavior could be improved.

Problem 1: Yeo-Johnson Transformation of Bond Volatility

The original AR(1) model for the logarithm of PC1 bond volatility was

$$\ln(V_t) = \alpha + \beta \ln(V_{t-1}) + Z_t.$$

The original residual diagnostics rejected normality based on both the Shapiro-Wilk and Jarque-Bera tests. Therefore, a Yeo-Johnson transformation was applied to $\ln(V_t)$.

The estimated Yeo-Johnson transformation parameter was

$$\lambda = -0.01259.$$

The transformed series was then used to estimate another AR(1) model. The estimated AR(1) coefficient was

$$\hat{\beta} = 0.8544.$$

The residuals from the transformed model were examined using normality tests, Ljung-Box tests, ACF plots, and a Q-Q plot.

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.1640	Fail to reject normality
Jarque-Bera	0.3434	Fail to reject normality
Ljung-Box Z_t, lag 5	0.03732	Reject no autocorrelation
Ljung-Box Z_t, lag 10	0.09814	Fail to reject

Diagnostic	p-value	Decision at 5%
Ljung-Box $\|Z_t\|$, lag 5	0.3735	Fail to reject
Ljung-Box $\|Z_t\|$, lag 10	0.7142	Fail to reject

The Yeo-Johnson transformation substantially improved the normality of the residuals. Both the Shapiro-Wilk and Jarque-Bera tests now fail to reject normality at the 5% significance level. In addition, there is no statistically significant autocorrelation in the absolute residuals.

However, the Ljung-Box test for the residuals through lag 5 remains significant, with a p-value of 0.03732.

Problem 2: Yeo-Johnson Transformation of the VAR Model

In Problem 2, a VAR(1) model was estimated using the logarithm of PC1 bond volatility and the logarithm of S&P volatility. The original diagnostics indicated that the bond-volatility innovation, Z_{1t}, was not consistent with an IID Gaussian process.

Yeo-Johnson transformations were therefore applied to both variables, and the VAR(1) model was re-estimated using the transformed series.

The residuals from the transformed VAR model are denoted by Z_{1t} for the bond-volatility equation and Z_{2t} for the S&P-volatility equation.

Bond Volatility Innovation, Z_{1t}

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.1951	Fail to reject normality
Jarque-Bera	0.4079	Fail to reject normality
Ljung-Box Z_{1t}, lag 5	0.04988	Reject no autocorrelation
Ljung-Box Z_{1t}, lag 10	0.09578	Fail to reject
Ljung-Box $\|Z_{1t}\|$, lag 5	0.4582	Fail to reject
Ljung-Box $\|Z_{1t}\|$, lag 10	0.7838	Fail to reject

For the bond-volatility innovation, the Yeo-Johnson transformation substantially improved normality. Both normality tests now fail to reject the null hypothesis of normality.

However, the lag-5 Ljung-Box test has a p-value of 0.04988, which is below the 5% significance level.

S&P Volatility Innovation, Z_{2t}

Diagnostic	p-value	Decision at 5%
Shapiro-Wilk	0.2976	Fail to reject normality
Jarque-Bera	0.7913	Fail to reject normality
Ljung-Box Z_{2t}, lag 5	0.7484	Fail to reject
Ljung-Box Z_{2t}, lag 10	0.8067	Fail to reject
Ljung-Box $\|Z_{2t}\|$, lag 5	0.8876	Fail to reject
Ljung-Box $\|Z_{2t}\|$, lag 10	0.8958	Fail to reject

The diagnostics for Z_{2t} are still consistent with IID Gaussian behavior.

Conclusion

The Yeo-Johnson transformation substantially improved the residual diagnostics for Problems 1 and 2. In particular, the normality problems observed for the bond-volatility innovations in the original models were no longer present after transformation.